

Agile Project Management

SECTION: Origins: The Agile Manifesto (2001)

Agile was formalized around four core values:

1. Individuals and interactions **over** processes and tools
2. Working software **over** comprehensive documentation
3. Customer collaboration **over** contract negotiation
4. Responding to change **over** following a plan

Note: the items on the right still have value — Agile just prioritizes the items on the left.

SECTION: Twelve Agile Principles (Key Ones for Exams)

- Highest priority is customer satisfaction through early and continuous delivery of valuable software.
- Welcome changing requirements, even late in development.
- Deliver working software frequently (weeks rather than months).
- Business people and developers must work together daily.
- Build projects around motivated individuals; give them the environment and support they need.
- Working software is the primary measure of progress.
- Simplicity — maximizing the amount of work *not* done — is essential.
- The best architectures/requirements/designs emerge from self-organizing teams.

SECTION: Scrum Framework

The most widely used Agile framework.

Roles:

- **Product Owner:** owns and prioritizes the product backlog, represents stakeholder/business interests.
- **Scrum Master:** facilitates the process, removes impediments, shields the team from disruption; not a traditional "manager."
- **Development Team:** self-organizing, cross-functional group that builds the increment.

Artifacts:

- **Product Backlog:** prioritized list of everything that might be needed in the product.
- **Sprint Backlog:** subset of the product backlog selected for the current sprint, plus a plan for delivering it.

- **Increment:** the sum of all completed product backlog items at the end of a sprint, meeting the "Definition of Done."

Events:

- **Sprint:** a fixed-length iteration, typically 1–4 weeks, producing a potentially shippable increment.
- **Sprint Planning:** team selects backlog items and plans the sprint.
- **Daily Scrum (Stand-up):** 15-minute daily sync (what I did, what I'll do, blockers).
- **Sprint Review:** demo the increment to stakeholders, gather feedback.
- **Sprint Retrospective:** team reflects on process and identifies improvements for the next sprint.

SECTION: Kanban

A visual workflow management method using a **Kanban board** with columns (e.g. To Do / In Progress / Done). Key principles: visualize work, limit **Work in Progress (WIP)**, manage flow, make process policies explicit, and improve collaboratively. Unlike Scrum, Kanban has no fixed iterations — it's a continuous flow system.

SECTION: Estimation Techniques

- **Story Points:** relative sizing of effort/complexity/uncertainty (often Fibonacci-like sequence: 1, 2, 3, 5, 8, 13).
- **Planning Poker:** team members privately estimate story points, then reveal simultaneously to surface and discuss differences in understanding.
- **Velocity:** the average number of story points a team completes per sprint, used to forecast future capacity.

SECTION: Agile vs Waterfall (Predictive) Comparison

Aspect	Waterfall	Agile
Requirements	Fixed upfront	Evolve iteratively
Delivery	One deliverable at end	Frequent incremental releases
Change	Discouraged/costly late	Welcomed throughout
Customer involvement	Mainly at start/end	Continuous
Documentation	Heavy, formal	Lightweight, "just enough"
Best suited for	Stable, well-understood requirements (e.g. construction, regulated systems)	Uncertain/evolving requirements (e.g. new software products)

SECTION: Common Exam Angles

- Explain a Scrum role/artifact/event and its purpose.
- Compare Agile vs Waterfall for a given project scenario and justify which is more suitable.

- Calculate/interpret velocity given past sprint data.
- Distinguish Scrum from Kanban.